

Evaluating the Temporal Robustness of Twitter Bot Detection Models

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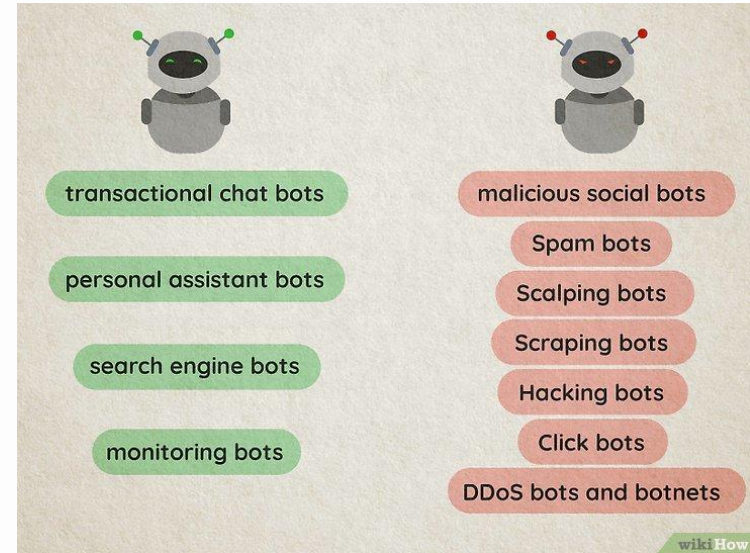
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Introduction

The Growing Threat of Social Media Bots

Motivation: Social Media Bots

- Accounts controlled programmatically to mimic human behavior and amplify messages at scale
- Not all bots are harmful (routine updates and services are legitimate uses)
- **Malicious bots:** spread spam, artificially amplify messages, promote disinformation, etc.



Real-World Impact

- **2016 U.S. Presidential Election:** bots amplified partisan content and distorted political discussion
- **2017 French Presidential Election:** bots ran disinformation campaigns to sway public engagement
- **Ongoing:** bots promote low-credibility information early in its spread, before corrections can catch up



The Challenge + Our Goal

- Newer bots are designed to closely simulate human behavior, fooling both humans and ML systems
- Detection improvements drive bot sophistication. This is an ongoing arms race.
- **Key gap:** cross-temporal generalization of detection models remains underexplored.

Our paper analyzes the performance of machine learning models to identify social media bots on Twitter (x.com) over time.

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Related Work

What has been done, and what's missing?

Prior Research

1. **Bot Impact:** Ferrara et al. (2016), Bessi & Ferrara (2016), Shao et al. (2018): bots manipulate elections and amplify misinformation
2. **Bot Evolution:** Cresci et al. (2017), Cresci (2020): bots increasingly evade detection; single-benchmark evaluation is insufficient
3. **Tools & Datasets:** Botometer (Yang et al., 2022); TwiBot-20 & TwiBot-22 (Feng et al.): prior SOTA models struggle on newer, harder data

What's Missing?

- High performance on one benchmark does not guarantee performance across time
- Few studies test whether supervised models transfer across datasets from different years
- Which features remain informative as bot behavior evolves is underexplored
- **Our paper:** evaluates 6 supervised models across 4 Twitter datasets spanning 2015-2022 to directly address this gap

Methodology

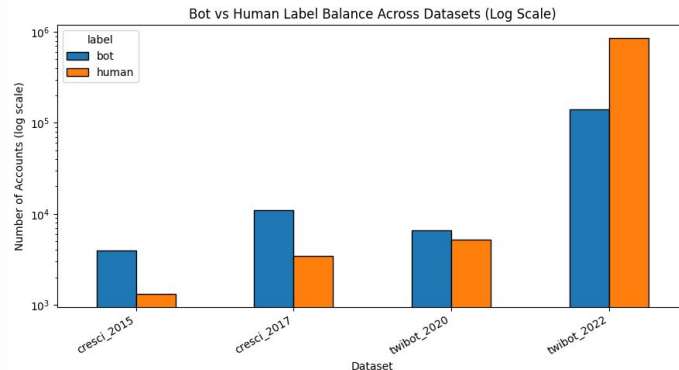
How We Tested Temporal Generalization

Datasets

- Cresci-15
 - Human Annotated
 - 3987 bots, 1314 humans, 2.8 million tweets
- Cresci-17
 - Human Annotated
 - 10894 bots, 3474 humans, over 6 million tweets
- Twibot-2020
 - Human Annotated (crowd sourced)
 - 6589 bots, 5237 humans, 2 million tweets
- Twibot-2022
 - Annotated by supervised classification models
 - 139,943 bots, 860,057 humans, 88,217,457 tweets

Exploratory Data Analysis

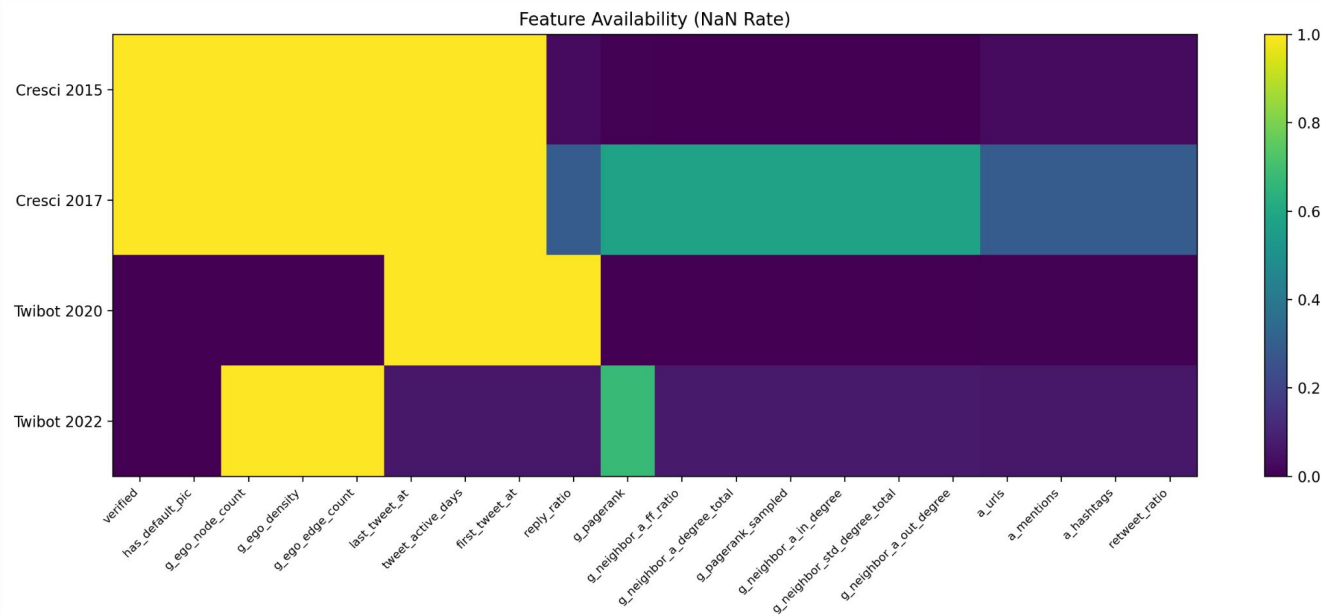
- Looking for missingness
- Percentage of features
- Ratios between classes
- Examples of accounts and tweets
- Better understanding of features and distributions in the datasets



	dataset	column	missing_count	missing_pct
9	cresci_2015	verified	5301	100.00
12	cresci_2015	default_profile_image	5276	99.53
10	cresci_2015	description	1216	22.94
11	cresci_2015	location	1212	22.86
24	cresci_2017	verified	14357	99.92
27	cresci_2017	default_profile_image	14290	99.46
26	cresci_2017	location	6684	46.52
25	cresci_2017	description	5744	39.98
56	twibot_2022	location	291564	29.16
55	twibot_2022	description	124262	12.43

Feature Engineering

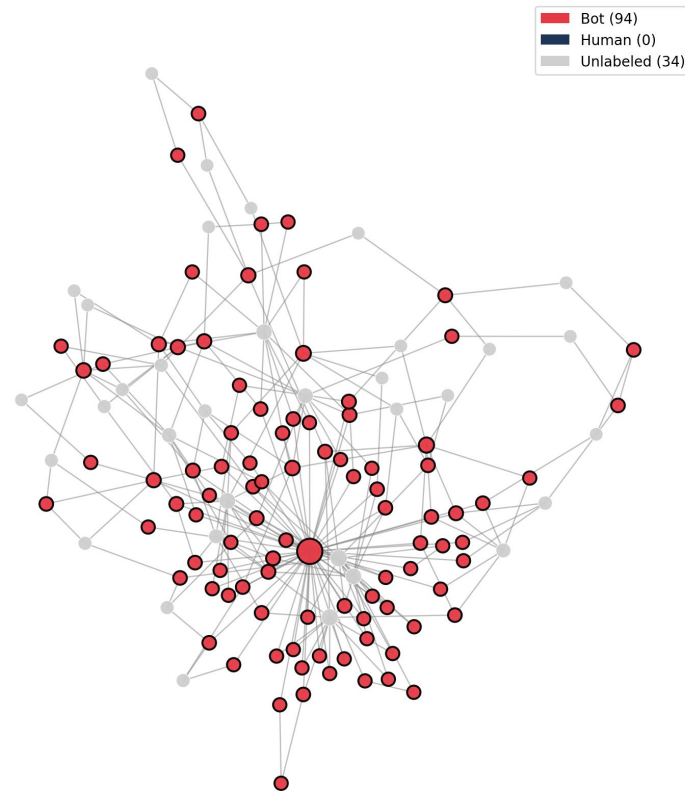
After conducting an Exploratory Data Analysis, we systematically selected features which might best detect differences in bot and human accounts.



Feature Engineering

- Core Account Features:
 - account age
 - follower/friend ratio
 - profile completeness
 - tweet count
 - retweet ratio
 - hashtag/mention/URL averages
 - text length averages
 - save base feature dataset
- Graph Features
 - Degree
 - Reciprocity
 - Component
 - Neighbor data
 - Page rank and Ego data (optional)

Cresci 2017 — connected neighborhood around a bot (128 nodes, 238 edges)



Models and Experiments

Machine Learning Models

- Logistic regression
- Ridge regression
- Lasso regression
- Random forest
- XGBoost
- Linear support vector machine

Each model was trained on one dataset and tested on a different dataset. This was repeated until every model model was trained and tested on every possible pairwise dataset combination. For each dataset pair only features available to both datasets were used.

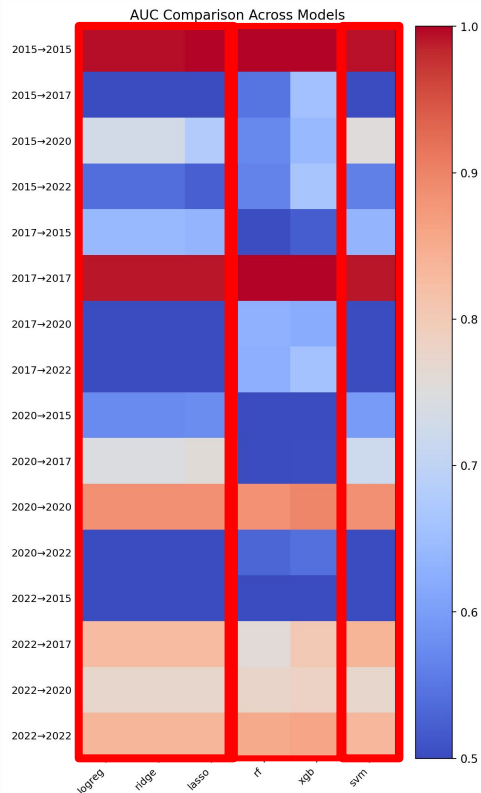
Afterwards we measured Area Under the Receiver Operating Characteristic Curve (AUC) which is a measure of how well a model separates positive and negative classes and AUC degradation between train and test datasets.

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Results

What data did we find?

Overall AUC Heatmap

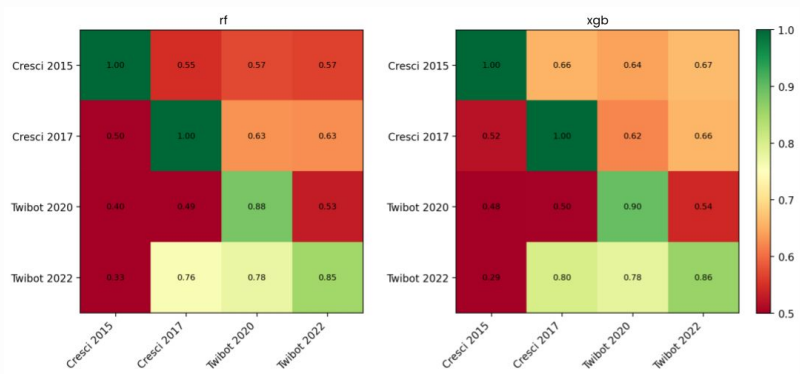


Key Observations:

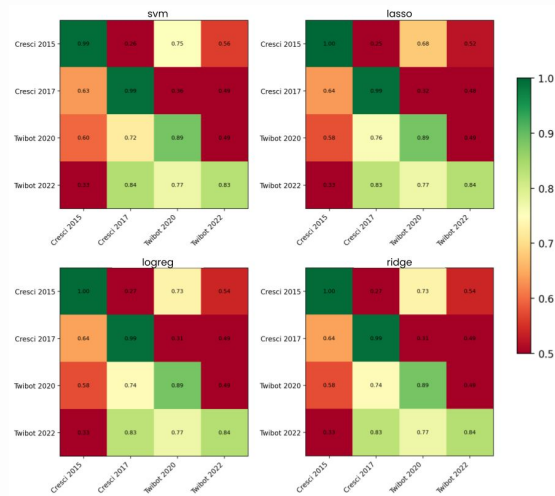
- Models trained and tested on the same dataset have consistently high scores (as expected)
- Mixed results for cross-era pairs
 - 2015→2020, 2020→2017, 2022→2017, and 2022→2020 perform well
 - Other pairs do not...
- There is a noticeable distinction between the linear models and the tree-based models...

Takeaway: We can generalize the data as a binary distinction between the two model families, where the choice of family matters more than the choice of a specific model.

More on this generalization...

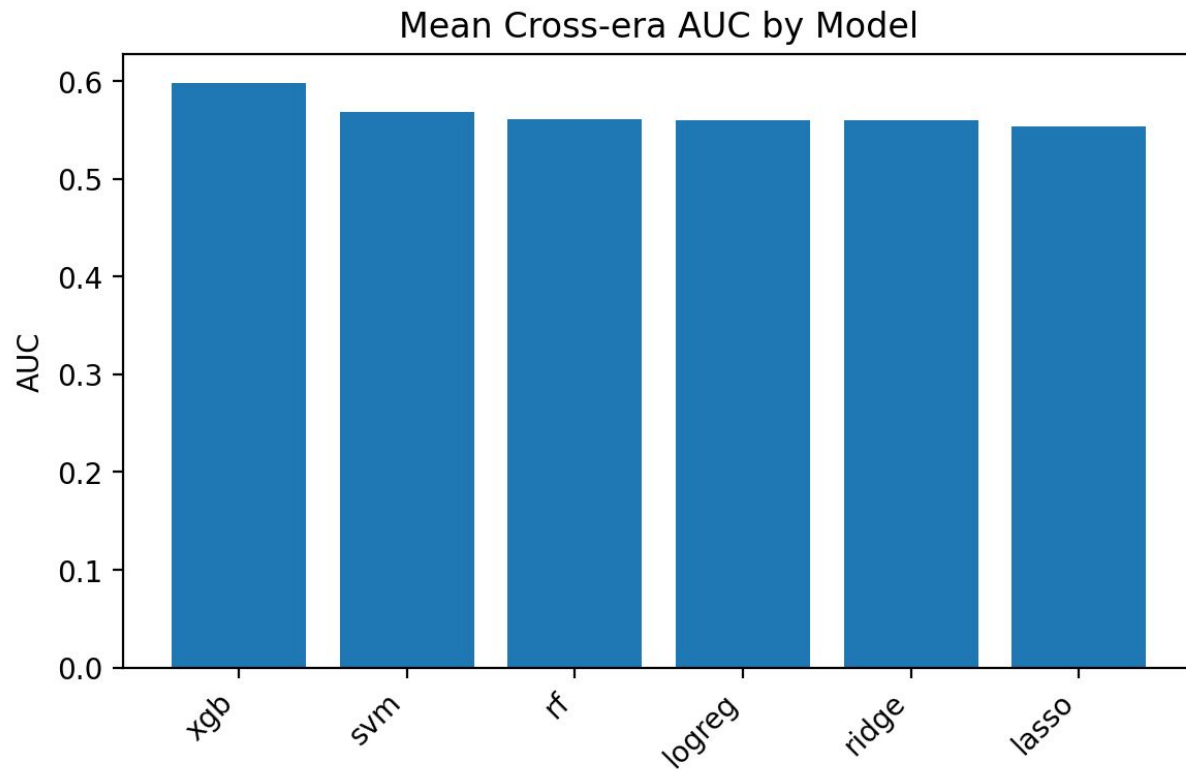


Tree-Based AUC Matrices



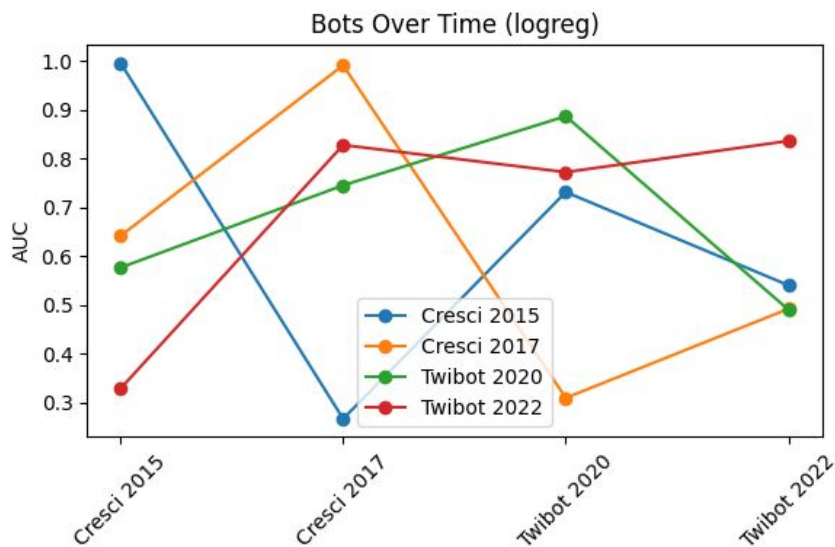
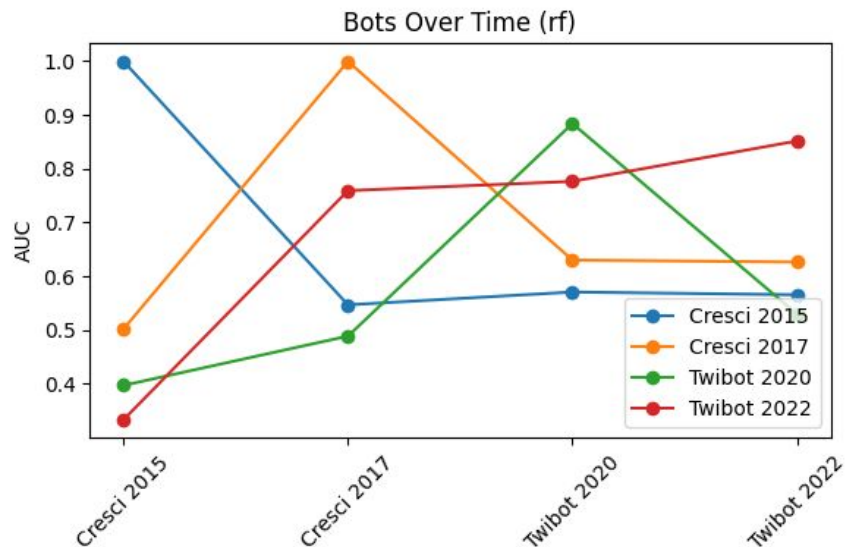
Linear AUC Matrices

Mean performance by Model



Takeaway: XGBoost did the best, but the range is small.

Key Metric: Bots Over Time



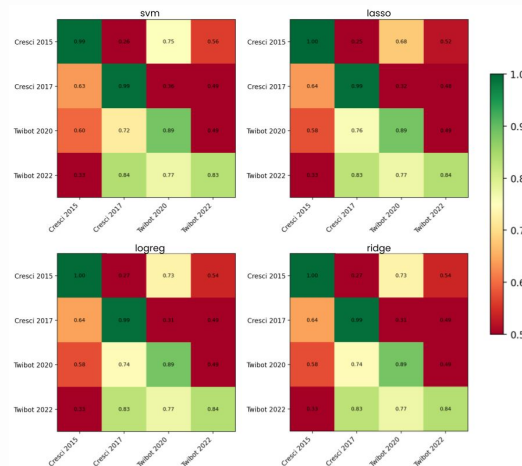
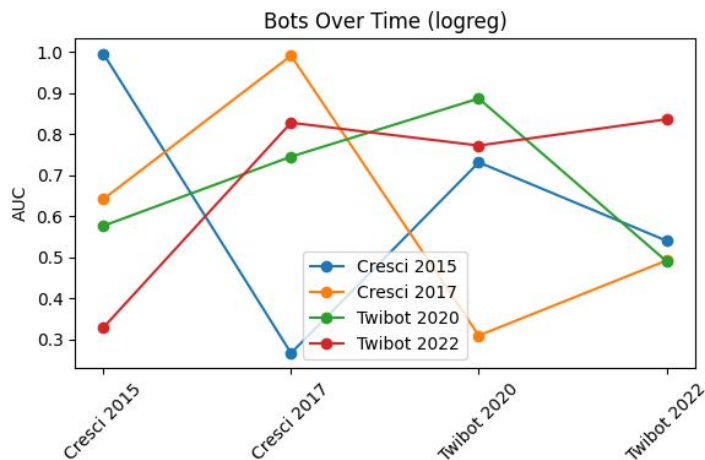
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Conclusion

What does this mean?

Conclusions from the Results

- There were a few key points that could have caused the discrepancy
- Data was likely homogenous, features were not as distinct, and datasets were not varied
- However, there did seem to be an overall trend that models trained on older datasets performed worse as time progressed



Future Steps

- Future research could investigate a wider range of features across the separate models
- Profile photo, distinguish on different types of bots, etc.
- If possible, a wider range of data across the years could be used, but this is expensive and difficult to obtain
- Similar research has been conducted on this matter. A investigation conducted by Sheetal and Kaur split bots into various categories such as Spam bots, social bots, cyborgs, but showcased additional dataset limitations. (Sheetal and Kaur, 2025).
- Another investigation conducted by Davoudi et al. scored models based on additional features such as mean post length, tweet diversity, user name, and profile photo (Davoudi et al., 2020).

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